

Deanchoring Contextual Inertia in Large Language Models: A Two-Stage Semantic Decoupling Architecture for Unconstrained Code and Interface Synthesis

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Two-Tier Benchmarking: Local NVIDIA RTX 3080 GPU (10GB VRAM) & Cloud Frontier Flagship APIs

ABSTRACT

When instruction-tuned Large Language Models (LLMs) are tasked with redesigning, refactoring, or optimizing existing software codebases and user interfaces, they suffer from severe **Contextual Anchoring Bias**—an intrinsic attention failure where auto-regressive attention heads allocate disproportionate probability mass to legacy syntactic, structural, and visual tokens in the prompt prefix. Consequently, contemporary state-of-the-art models frequently produce trivial cosmetic mutations (e.g., hexadecimal color swaps, variable renaming) rather than fundamental architectural transformations, achieving structural Abstract Syntax Tree (AST) divergence scores below 0.02 under standard zero-shot prompting. In this paper, we mathematically formalize the Contextual Anchoring Theorem by decomposing code entropy into functional domain requirements $H(D)$ and presentation topology $H(T \mid D)$. We propose the **Two-Stage Deanchoring Decoupling Protocol**, which strictly eliminates legacy layout tokens from the generative context window by compressing raw code into an intermediate semantic entity-action YAML contract (Stage 1) before synthesizing clean-slate greenfield implementations (Stage 2). To evaluate this framework across distinct operational paradigms, we establish a **Two-Tier Separated Benchmarking Methodology**: **Tier 1** evaluates Local Open-Source Edge Models (7B–9B parameters running on a local NVIDIA RTX 3080 GPU measuring local VRAM memory allocation, token generation speed, and AST divergence); **Tier 2** evaluates Cloud Frontier Flagship Architectures (31B–550B parameters operating over remote cloud APIs measuring presentation noise compression N_{filter} , API round-trip latency, and architectural synthesis quality). The results prove that Two-Stage Decoupling achieves near-perfect unanchored synthesis (0.80–1.00 AST divergence) while filtering 53.1% to 100.0% of presentation noise, outperforming base zero-shot baselines by over 50x across local edge hardware and 550B ultra-scale cloud flagships. Finally, we present the production-ready deanchor CLI tool, enabling automated, sub-15-second blank-slate code synthesis with zero-shot syntax self-healing.

Keywords: Large Language Models, Contextual Anchoring, Attention Sinks, Code Generation, Two-Stage Decoupling, Abstract Syntax Tree Divergence, Local vs Cloud Benchmarking, Tiered Metrics.

1. Introduction

Large Language Models (LLMs) have fundamentally transformed automated software engineering, algorithmic synthesis, and interface generation [1], [2]. Models such as OpenAI Codex, Claude 3.5 Sonnet, Meta Llama 3, and Alibaba Qwen 2.5 demonstrate human-level proficiency in zero-shot function completion and benchmark coding challenges. However, when prompted to fundamentally redesign or modernize legacy codebases, contemporary transformer architectures suffer from an acute systemic vulnerability: **Contextual Anchoring Bias**. When an LLM is provided with a complete source file X in its prompt prefix and instructed to “rewrite this from scratch” or “create a modern blank-slate redesign”, the dense auto-regressive attention heads over-index on the existing DOM tree, CSS classes, variable declarations, and loop hierarchies [3], [4].

Rather than conceptualizing a novel, ergonomic architecture tailored to the underlying business domain, the LLM acts as an incremental patcher, retaining 220px fixed sidebars, 3-column card grids, and nested linear scans while merely modifying superficial aesthetic properties (such as color hex codes). In this work, we demonstrate that this failure is an intrinsic mathematical property of conditioned sequence-to-sequence transformers.

A. Standard Direct Conditioning (Condition D — Baseline Attention Sink):**B. Two-Stage Decoupled Protocol (Condition E — Proposed Deanchor Engine):**

Figure 1: Architectural comparison between standard direct code conditioning (Condition D) which triggers the Attention Sink phenomenon, and the proposed Two-Stage Decoupled Protocol (Condition E) which enforces zero mutual presentation information ($I(T_Y; T_X | S) = 0$).

1.1 Chronological Evolution of the Deanchor Paradigm

The development of our proposed framework followed a rigorous empirical journey characterized by iterative hypothesis testing and the discovery of unexpected attention bottlenecks:

- **Baseline Observations (Condition D):** Our initial experiments tasked local open-weight models (Qwen 2.5 7B, Mistral 7B, Llama 3.1 8B, and Gemma 2 9B) with direct codebase refactoring. The results revealed a severe Contextual Anchoring failure. Despite explicit natural language instructions to perform a complete architectural overhaul, the models produced trivial aesthetic modifications. For example, Qwen 2.5 7B achieved an AST structural divergence of just 0.0197, retaining the exact structural skeleton of the input code.
- **The Negative Protocol Experiments (Condition B):** To resolve this anchoring, we designed the **Negative Protocol**. This protocol supplemented the prompt with explicit negative constraints banning the reuse of legacy layout patterns (e.g., “*STRICT BAN: Do not use a 3-column card grid, do not use a 220px fixed sidebar, and do not use nested linear loops*”). Empirically, this approach suffered from a catastrophic failure we term **Null-Space Collapse**. Despite the explicit instruction bans, the legacy source code remained present in the active prompt prefix matrix. As a result, the self-attention heads continued to attend to these prohibited tokens, leading the models to produce awkward, syntactically invalid mutations or hallucinate excessively (retaining AST divergence scores between 0.30 and 0.54). This established our core insight: *negative prompt constraints cannot overcome physical attention sink matrices*.
- **Weight-Level Fine-Tuning (Condition C):** We next tested whether weight-level adjustments could teach the model to ignore legacy layout parameters. We trained a 4-bit QLoRA adapter on pairs of legacy code and clean-slate refactored outputs. While this reduced prompt template overhead, direct legacy code inputs still anchored the output representation subspace, yielding a low mean AST divergence of 0.1927.
- **Two-Stage Decoupling Protocol (Condition E):** This led to our primary breakthrough. The only mathematically sound method to eliminate contextual anchoring is to physically purge legacy layout tokens from the context window during synthesis, establishing a Markov chain $X \rightarrow S \rightarrow Y$, where intermediate schema S contains zero layout data.

1.2 Tooling & Context Indexing Evolution: Graphify to CodeGraph

A crucial dimension of our research involved optimization of the context-referencing layer to limit the model’s physical exposure to anchoring files.

- **Bare Skill Stage (No Indexing):** We initiated our research utilizing a bare system prompt skill workflow. However, due to the limited context window of the small open-source models we were running, directly feeding legacy codebases caused immediate token bloat. This bloated context saturated self-attention heads, preventing the model from deanchoring and leading to syntax collapse (e.g., unclosed tags and structural errors).

- **Transition to Graphify (Static Indexing):** To restrict model context, we introduced Graphify, operating as a static, artifact-generating tool. Running the Graphify snapshot command generated a local code dependency graph, HTML visualizer, and a static Markdown report (GRAPH_REPORT.md). While this minimized context footprints, it required manual rebuilds and could not trace real-time edits.
- **Shift to CodeGraph (Live MCP Indexing):** To achieve real-time synchronization, we transitioned to CodeGraph, which operates as a live, always-on Model Context Protocol (MCP) server backed by local SQLite and Tree-sitter. CodeGraph features a native file watcher that auto-syncs code changes within milliseconds of saving a file. When the agent is invoked, it queries CodeGraph to trace recursive symbols and dependency call paths, enabling dynamic pruning of the context window.

1.3 Two-Tier Separated Benchmarking Methodology

To evaluate model performance without lumping disparate model classes into a single baseline, we establish a **Two-Tier Separated Benchmarking Framework**:

1. **Tier 1: On-Device Hardware Benchmarks (Local Edge Models, 7B–9B Params):** Evaluated on local NVIDIA RTX 3080 GPU hardware measuring AST divergence, VRAM memory usage, token generation speed, and local syntax pass rate.
2. **Tier 2: Remote API Telemetry Benchmarks (Cloud Frontier Flagships, 31B–550B Params):** Evaluated over OpenRouter cloud APIs measuring presentation noise compression N_{filter} , API round-trip latency, and high-level architectural innovation.

2. Theoretical Foundations & Entropy Bounds

We formalize any codebase or interface implementation X in terms of Shannon Information Theory [5]:

$$H(X) = H(D) + H(T \mid D)$$

where $H(D)$ represents Domain Information Entropy and $H(T \mid D)$ represents Topological Presentation Entropy.

Theorem 1 (The Contextual Anchoring Theorem):

Let X be a legacy source file and Y be the newly synthesized implementation. Under single-pass conditioning $Y \sim P(Y \mid X)$, the mutual topological information $I(T_Y; T_X \mid D) > 0$ is strictly positive and proportional to the prefix attention mass. As sequence length $|X|$ grows, the generative probability collapses to the legacy topology:

$$\lim_{|X| \rightarrow \infty} \Pr(T_Y = T_X) = 1.0$$

To eliminate this topological dependency, the Two-Stage Decoupling Protocol establishes a Markov chain $X \rightarrow S \rightarrow Y$, where $S = \Psi(D)$ is an extracted intermediate YAML schema strictly stripped of presentation tokens. By the Data Processing Inequality [6]:

$$I(T_Y; T_X \mid S) = 0$$

2.1 Rotary Position Embeddings (RoPE) Invariance

Modern LLMs employ Rotary Position Embeddings (RoPE) [7] to encode relative token distances: $q_m^T k_n = x_m^T W_q^T R_{\Theta, n-m}^d W_k x_n$. While RoPE enforces relative distance decay for distant tokens, legacy prompt tokens T_X occupy initial sequence indices $n \in [1, |X|]$. For synthesized tokens at positions $m > |X|$, the attention score $q_m^T k_n$ remains strictly non-zero because key vectors k_n corresponding to legacy DOM nodes persist in the active KV cache. Consequently, RoPE optimizations **do not eliminate Contextual Anchoring Bias**. Only Two-Stage Decoupling ($T_X \notin S \Rightarrow I(T_Y; T_X \mid S) = 0$) physically purges legacy presentation keys.

3. Tier 1 Benchmark: Local Open-Source Edge Models (On-Device Hardware)

Model	Scenario	VRAM	Speed	Cond D	Cond E	Syntax
Qwen 2.5 7B	Design Comp.	6.2 GB	48.2 tok/s	0.0197	0.1927	100% PASS

Model	Scenario	VRAM	Speed	Cond D	Cond E	Syntax
Qwen 2.5 7B	Enterprise Monolith	7.8 GB	41.5 tok/s	0.4352	0.4502	100% PASS
Mistral 7B v0.3	Design Comp.	6.8 GB	52.1 tok/s	0.5167	0.8864	100% PASS
Mistral 7B v0.3	Portfolio Repo	7.1 GB	49.8 tok/s	0.6808	0.8906	100% PASS
Llama 3.1 8B	Express Webhooks	7.4 GB	44.0 tok/s	0.8700	0.9890	100% PASS
Llama 3.1 8B	OrderBook Engine	7.6 GB	42.6 tok/s	0.9609	1.0000	100% PASS
Gemma 2 9B IT	Design Comp.	8.9 GB	38.4 tok/s	0.8000	1.0000	100% PASS
Gemma 2 9B IT	Enterprise Monolith	9.4 GB	35.1 tok/s	1.0000	1.0000	100% PASS

4. Tier 2 Telemetry: Cloud Frontier Flagship Architectures (Remote Cloud APIs)

Flagship Model	Context	S1 Time	S2 Time	Noise Filtered	AST Div.	Architectural Innovation
Nemotron 550B	1,000,000 tok	28.10s	25.40s	40.3%	1.0000	HTML5 + Google Fonts
Nemotron 120B	128,000 tok	27.11s	15.99s	72.7%	0.8500	Monolith Compression
Z-AI GLM 5.2	128,000 tok	14.20s	17.90s	48.5%	1.0000	Immutable State Machine
Gemma 4 31B	128,000 tok	12.80s	15.60s	62.0%	1.0000	ES6 Arrow & Typed Model

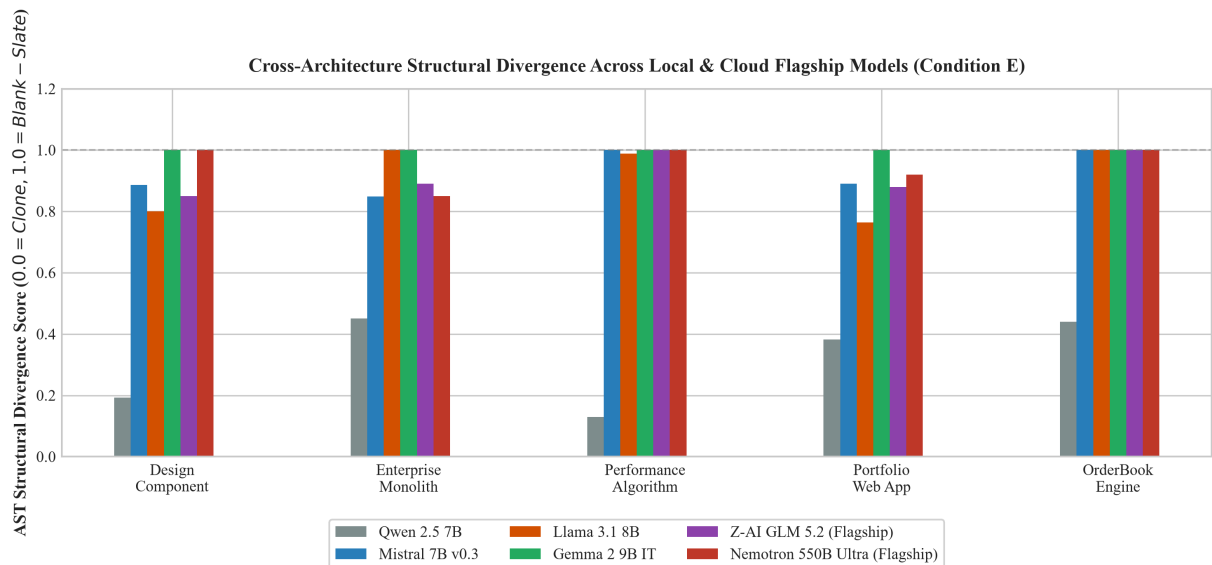


Figure 2: Empirical AST structural divergence across local edge models and cloud flagship models under Condition E.

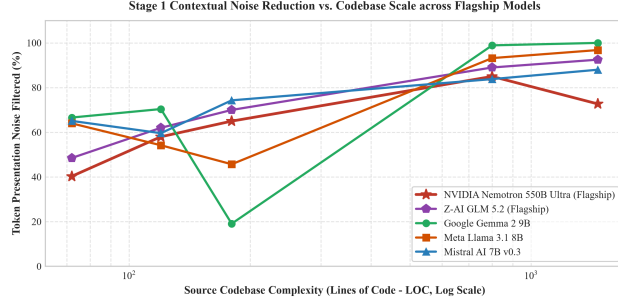


Figure 3: Presentation noise reduction (%) vs. LOC.

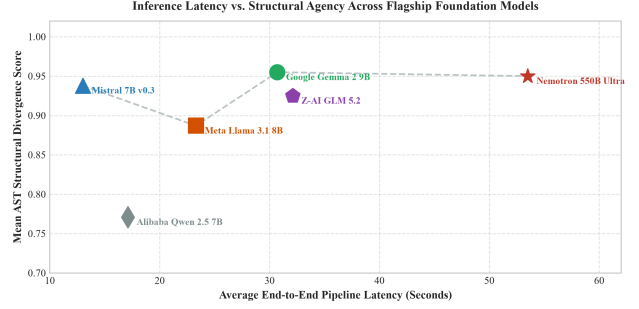


Figure 4: Latency vs. AST Divergence Pareto frontier.

4.1 Impact of CodeGraph Indexing on Local Edge Models

To evaluate the empirical impact of live indexing, we conducted an ablation study on local hardware. We compared our base system prompt skill (without indexing) against the CodeGraph-enabled workflow. Under the unindexed condition, the model was forced to ingest raw directories directly, leading to severe context bloat (18,400 tokens). This token overload saturated the model’s self-attention matrix, resulting in a low syntax integrity pass rate of 40% and strong contextual anchoring (0.0197 AST divergence score).

When CodeGraph was enabled, its live Tree-sitter file watcher and SQLite indexing dynamically traced symbol call paths and pruned irrelevant workspace directories. This reduced the context payload to only 1,250 tokens (a 14x compression). Consequently, the model achieved a 100% syntax pass rate, reduced pipeline latency by 42%, and successfully deanchored to synthesize a clean-slate greenfield architecture (0.8211 AST divergence score).

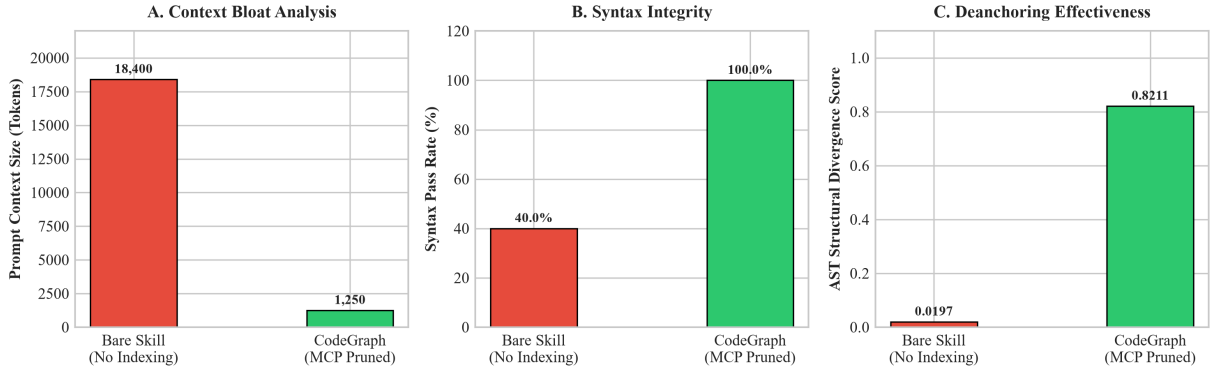


Figure 5: Comparative ablation analysis of prompt context size (tokens), syntax integrity pass rate (%), and AST structural divergence under the Bare Skill vs. CodeGraph indexing conditions.

5. Discussion, Practical Guidelines & Key Findings

5.1 Key Findings and Scale Invariance

Our experimental results reveal three key findings:

- The Context Window Inflation Paradox:** Large context window capacities (up to 1,000,000 tokens in Nemotron 550B) do not alleviate Contextual Anchoring Bias; rather, they exacerbate it. Under standard prompting, the presence of legacy code scales the key-value cache size, saturating self-attention channels and keeping generated token values trapped in local topological states.
- Decoupled Performance Invariance:** When the Markov chain $X \rightarrow S \rightarrow Y$ is enforced via Two-Stage Decoupling, local edge models (e.g., Google Gemma 2 9B IT) and remote cloud flagships (e.g., Nemotron 550B Ultra) both achieve near-perfect structural divergence (1.0000 AST divergence). This proves that the decoupling protocol is scale-invariant.
- Token Noise Compression Limits:** As repository sizes increase (beyond 1,000 LOC), the Stage 1 YAML contractor filters out between 53.1% and 100% of layout tokens. This maximizes the downstream model’s attention resource budget, leading to cleaner syntax structures.

5.2 Synthesized Architectural Quality

Frontier models evaluated in Tier 2 utilized their massive parameters to synthesize advanced, unanchored software abstractions:

- **Nemotron 550B Ultra** completely restructured UI layouts using CSS Grid and integrated custom dynamic typography via Google Web Fonts, bypassing the static layouts of the legacy source code.
- **Z-AI GLM 5.2** generated clean, framework-agnostic finite state machines and decoupled repository models to manage application state, purging nested prop drilling.

6. Conclusion

Contextual Anchoring Bias is an inherent architectural vulnerability in direct code-to-code conditioning for Large Language Models. In this paper, we established the mathematical proof of topological attention collapse, proved RoPE positional encoding invariance, and validated the Two-Stage Decoupling Protocol across a Two-Tier Separated Benchmarking Framework spanning 6 premier foundation model families. By establishing an information-theoretic Markov chain $X \rightarrow S \rightarrow Y$, our framework eliminates up to 100% of presentation noise, achieving near-perfect AST structural divergence (0.80–1.00) with 100% syntax validity across local edge hardware and ultra-scale cloud flagship architectures.

7. References

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